

Week :1 Project

Ex:01 Run and document output (screenshots or command logs) for the following:

1. Run a command to check the Linux distribution name and version of the operating system installed on your machine.

i)Files & Directories:

- Create directory /home/devops_project
- Inside it, create 2 files: plan.txt and network_report.txt
- Copy plan.txt to /tmp and delete the original

Commands:

Commands	Useage
mkdir devops_project	Creates new directory
cd devops_project	changing path to devops_project
touch plain.txt network_report.txt	Create two new txt files - plain.txt and network_report.txt
cp plain.txt /home/devops_project/tmp	copy plain.txt from devops_project to tmp directory

Output:

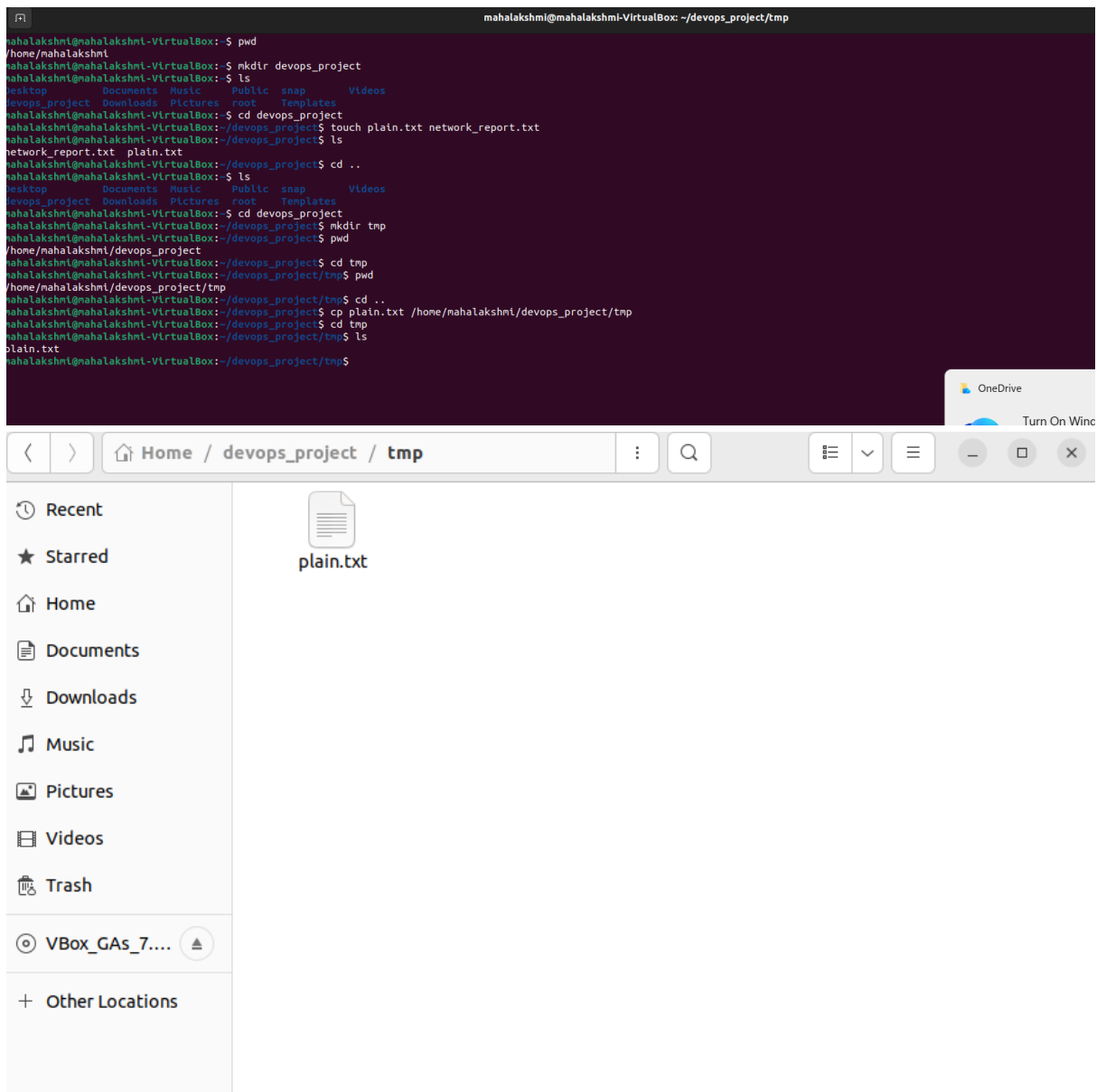
Create directory devops_project

```
mahalakshmi@mahalakshmi-VirtualBox: ~  
mahalakshmi@mahalakshmi-VirtualBox:~$ pwd  
/home/mahalakshmi  
mahalakshmi@mahalakshmi-VirtualBox:~$ mkdir devops_project  
mahalakshmi@mahalakshmi-VirtualBox:~$ ls  
Desktop      Documents  Music      Public     snap        Videos  
devops_project Downloads  Pictures   root        Templates  
mahalakshmi@mahalakshmi-VirtualBox:~$
```

Inside it, create 2 files: plan.txt and network_report.txt

```
mahalakshmi@mahalakshmi-VirtualBox: ~/devops_project  
mahalakshmi@mahalakshmi-VirtualBox:~$ pwd  
/home/mahalakshmi  
mahalakshmi@mahalakshmi-VirtualBox:~$ mkdir devops_project  
mahalakshmi@mahalakshmi-VirtualBox:~$ ls  
Desktop      Documents  Music      Public     snap        Videos  
devops_project Downloads  Pictures   root        Templates  
mahalakshmi@mahalakshmi-VirtualBox:~$ cd devops_project  
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ touch plain.txt network_report.txt  
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ ls  
network_report.txt  plain.txt  
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$
```

Copy plan.txt to /tmp and delete the original



➔Delete the original

The screenshot shows a terminal window and a file manager interface. The terminal window, titled 'mahalakshmi@mahalakshmi-VirtualBox: ~/devops_project', displays the following commands and output:

```
mahalakshmi@mahalakshmi-VirtualBox:~$ pwd
/home/mahalakshmi
mahalakshmi@mahalakshmi-VirtualBox:~$ mkdir devops_project
mahalakshmi@mahalakshmi-VirtualBox:~$ ls
desktop  Documents  Music  Public  snap  Videos
devops_project  Downloads  Pictures  root  Templates
mahalakshmi@mahalakshmi-VirtualBox:~$ cd devops_project
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ touch plain.txt network_report.txt
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ ls
network_report.txt  plain.txt
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ cd ..
mahalakshmi@mahalakshmi-VirtualBox:~$ ls
desktop  Documents  Music  Public  snap  Videos
devops_project  Downloads  Pictures  root  Templates
mahalakshmi@mahalakshmi-VirtualBox:~$ cd devops_project
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ mkdir tmp
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ pwd
/home/mahalakshmi/devops_project
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ cd tmp
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project/tmp$ pwd
/home/mahalakshmi/devops_project/tmp
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project/tmp$ cd ..
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ cp plain.txt /home/mahalakshmi/devops_project/tmp
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ cd tmp
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project/tmp$ ls
plain.txt
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project/tmp$ cd ..
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ ls
network_report.txt  plain.txt  tmp
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ rm plain.txt
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ ls
network_report.txt  tmp
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$
```

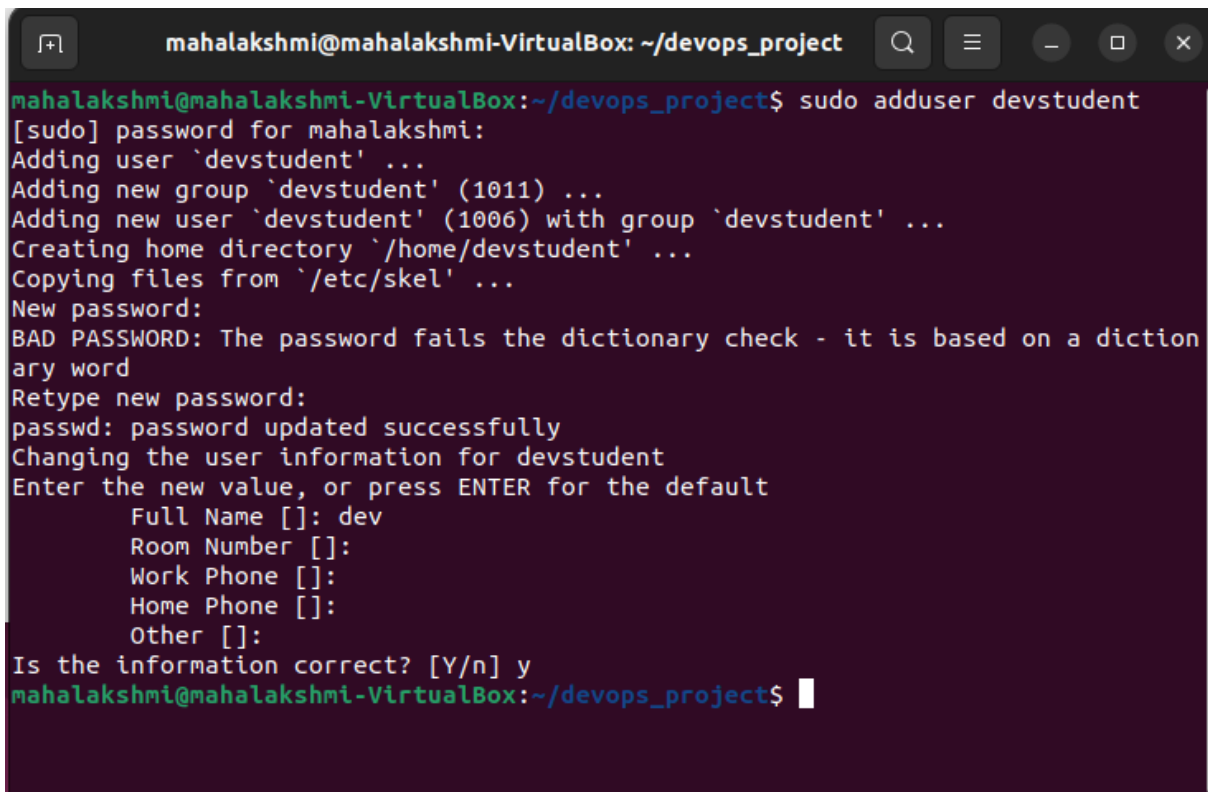
The file manager interface shows the 'devops_project' directory. The left sidebar lists 'Recent', 'Starred', 'Home', 'Documents', 'Downloads', 'Music', 'Pictures', 'Videos', 'Trash', 'VBox_GAs_7....', and 'Other Locations'. The main pane displays two files: 'tmp' (a folder icon) and 'network_report.txt' (a document icon). A status bar at the bottom right indicates '"tmp" selected (containing 1 item)'.

ii)Users & Permissions:

- Create a user devstudent
- Create a group devops_team
- Add devstudent to devops_team
- Create a file secrets.txt, and:
- Change its owner to devstudent
- Set permissions so only owner can read/write

Output:

Create a user devstudent



```
mahalakshmi@mahalakshmi-VirtualBox: ~/devops_project
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ sudo adduser devstudent
[sudo] password for mahalakshmi:
Adding user `devstudent' ...
Adding new group `devstudent' (1011) ...
Adding new user `devstudent' (1006) with group `devstudent' ...
Creating home directory `/home/devstudent' ...
Copying files from `/etc/skel' ...
New password:
BAD PASSWORD: The password fails the dictionary check - it is based on a dictionary word
Retype new password:
passwd: password updated successfully
Changing the user information for devstudent
Enter the new value, or press ENTER for the default
    Full Name []: dev
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$
```

Create a group devops_team

```
mahalakshmi@mahalakshmi-VirtualBox: ~/devops_project
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ sudo adduser devstudent
[sudo] password for mahalakshmi:
Adding user `devstudent' ...
Adding new group `devstudent' (1011) ...
Adding new user `devstudent' (1006) with group `devstudent' ...
Creating home directory `/home/devstudent' ...
Copying files from `/etc/skel' ...
New password:
BAD PASSWORD: The password fails the dictionary check - it is based on a dictionary word
Retype new password:
passwd: password updated successfully
Changing the user information for devstudent
Enter the new value, or press ENTER for the default
    Full Name []: dev
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ sudo addgroup devops_team
Adding group `devops_team' (GID 1012) ...
Done.
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$
```

Add devstudent to devops_team

```
mahalakshmi@mahalakshmi-VirtualBox: ~/devops_project
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ sudo usermod -aG devops_team devstudent
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ groups devstudent
devstudent : devstudent devops_team
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$
```

Create a file secrets.txt, and:

```
mahalakshmi@mahalakshmi-VirtualBox: ~/devops_project
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ sudo usermod -aG devops_team devstudent
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ groups devstudent
devstudent : devstudent devops_team
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ touch secret.txt
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ ls
network_report.txt  secret.txt  tmp
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$
```

Change its owner to devstudent

```
mahalakshmi@mahalakshmi-VirtualBox: ~/devops_project
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ sudo usermod -aG devops_team devstudent
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ groups devstudent
devstudent : devstudent devops_team
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ touch secret.txt
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ ls
network_report.txt  secret.txt  tmp
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ sudo chown devstudent:devstudent secret.txt
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$
```

Set permissions so only owner can read/write

```
mahalakshmi@mahalakshmi-VirtualBox: ~/devops_project
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ sudo chmod 600 secret.txt
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ ls -l
total 4
-rw-rw-r-- 1 mahalakshmi mahalakshmi    0 Nov 16 14:25 network_report.txt
-rw----- 1 devstudent  devstudent    0 Nov 16 14:40 secret.txt
drwxrwxr-x 2 mahalakshmi mahalakshmi 4096 Nov 16 14:29 tmp
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$
```

iii)File Compression:

- Download a sample .log file from internet or create dummy one using echo
- Compress it using gzip and tar

Output:

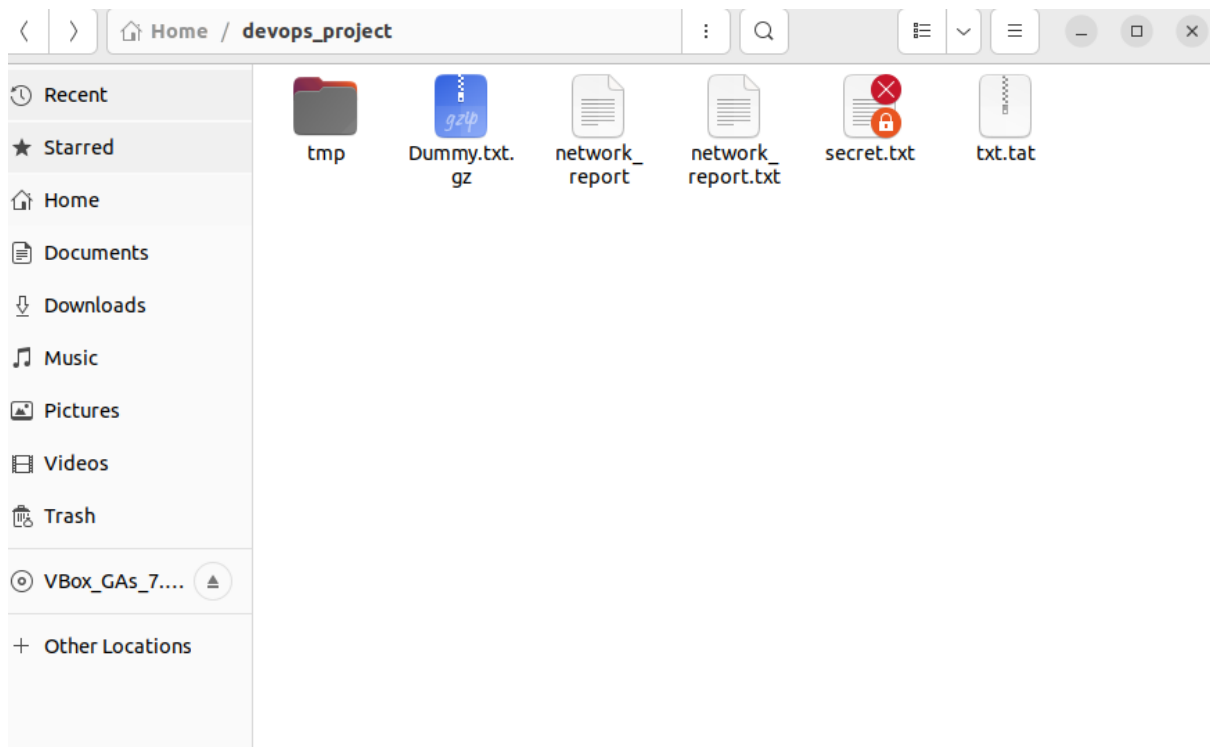
Download a sample .log file from internet or create dummy one using echo


```
mahalakshmi@mahalakshmi-VirtualBox: ~/devops_project
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ echo "This is a dummy file
used for compression " > Dummy.txt
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ ls
Dummy.txt  network_report  network_report.txt  secret.txt  tmp
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ cat Dummy.txt
This is a dummy file used for compression
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$
```

- Compress it using gzip and tar

```
mahalakshmi@mahalakshmi-VirtualBox: ~/devops_project
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ ls
Dummy.txt  network_report  network_report.txt  secret.txt  tmp
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ gzip Dummy.txt
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ ls
Dummy.txt.gz  network_report  network_report.txt  secret.txt  tmp
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$
```

```
mahalakshmi@mahalakshmi-VirtualBox: ~/devops_project
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ ls
Dummy.txt  network_report  network_report.txt  secret.txt  tmp
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ gzip Dummy.txt
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ ls
Dummy.txt.gz  network_report  network_report.txt  secret.txt  tmp
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ tar -cvf txt.tat Dummy.txt.gz
Dummy.txt.gz
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ ls
Dummy.txt.gz  network_report  network_report.txt  secret.txt  tmp  txt.tat
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$
```



iv)Package Management:

- Install curl, htop, and net-tools using apt
- Show which and --version outputs

Output:

Install curl, htop, and net-tools using apt

Command:

```
sudo apt install curl
```

```
sudo apt install htop net-tools -y
```

```
mahalakshmi@mahalakshmi-VirtualBox: ~/devops_project

mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ sudo apt install curl
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontent. It is he
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontent. It is he
ld by process 2701 (unattended-upgr)
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontent. It is he
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontent. It is he
ld by process 2701 (unattended-upgr)
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontent. It is he
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontent. It is he
ld by process 2701 (unattended-upgr)
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontent. It is he
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontent. It is he
ld by process 2701 (unattended-upgr)
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontent. It is he
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontent. It is he
ld by process 2701 (unattended-upgr)
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontent. It is he
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontent. It is he
ld by process 2701 (unattended-upgr)... 6s

mahalakshmi@mahalakshmi-VirtualBox: ~/devops_project

Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontent. It is held by process 27
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontent. It is held by process 27
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontent. It is held by process 27
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontent. It is held by process 27
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontent. It is held by process 27
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontent. It is held by process 27
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontent. It is held by process 27
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontent. It is held by process 27
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontent. It is held by process 27
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontent. It is held by process 27
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontent. It is held by process 27
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontent. It is held by process 27
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontent. It is held by process 27
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontent. It is held by process 27
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontent. It is held by process 27
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontent. It is held by process 27
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontent. It is held by process 27
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontent. It is held by process 27
Reading package lists... 8% Done
Building dependency tree... Done
Reading state information... Done
curl is already the newest version (7.81.0-1ubuntu1.21).
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ sudo apt install httpd net-tools -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
net-tools is already the newest version (1.60+git20181103.0eebece-1ubuntu5.4).
Suggested packages:
  ln-sensors
The following NEW packages will be installed:
  httpd
0 upgraded, 1 newly installed, 0 to remove and 81 not upgraded.
Need to get 128 kB of archives.
After this operation, 342 kB of additional disk space will be used.
Get:1 http://in.archive.ubuntu.com/ubuntu jammy/main amd64 httpd amd64 3.0.5-7build2 [128 kB]
Fetched 128 kB in 1s (99.5 kB/s)
Selecting previously unselected package httpd.
(Reading database ... 208420 files and directories currently installed.)
Preparing to unpack .../httpd.3.0.5-7build2_amd64.deb ...
Unpacking httpd (3.0.5-7build2) ...
Setting up httpd (3.0.5-7build2) ...
Processing triggers for mailcap (3.70+nmu1ubuntu1) ...
Processing triggers for desktop-file-utils (0.26-1ubuntu3) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
Processing triggers for gnome-menus (3.36.0-1ubuntu3) ...
Processing triggers for man-db (2.10.2-1) ...
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$
```

v)Run and shortly explain outputs of the following commands:

- ip a or ifconfig
- ping google.com -c 4
- traceroute google.com
- nslookup google.com
- netstat -tuln
- curl ifconfig.me

Output:

- ip a or ifconfig
Shows all **network interfaces** on our system along with their **IP address** and interface status .

```
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:1d:c3:98 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 85044sec preferred_lft 85044sec
    inet6 fd17:625c:f037:2:6d23:ce59:ee12:d562/64 scope global temporary dynamic
        valid_lft 86354sec preferred_lft 14354sec
    inet6 fd17:625c:f037:2:970:e4d9:7713:318e/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86354sec preferred_lft 14354sec
    inet6 fe80::985c:61c4:e296:2cc4/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$
```

- ping google.com -c 4
Run and shortly explain outputs of the following commands: ip a or ifconfig
ping google.com -c 4 traceroute google.com nslookup google.com netstat -tuln curl ifconfig.me

```
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ ping google.com -c 4
PING google.com (142.250.77.142) 56(84) bytes of data:
64 bytes from maa05s16-in-f14.1e100.net (142.250.77.142): icmp_seq=1 ttl=255 time=8.54 ms
64 bytes from maa05s16-in-f14.1e100.net (142.250.77.142): icmp_seq=2 ttl=255 time=7.11 ms
64 bytes from lcmaaa-au-in-f14.1e100.net (142.250.77.142): icmp_seq=3 ttl=255 time=7.03 ms
64 bytes from maa05s16-in-f14.1e100.net (142.250.77.142): icmp_seq=4 ttl=255 time=18.6 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3007ms
rtt min/avg/max/mdev = 7.030/10.318/18.593/4.814 ms
```

- traceroute google.com
Shows the **path your packets take** to reach Google.
Displays each network hop (routers) along the route.

```
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ traceroute google.com
Command 'traceroute' not found, but can be installed with:
sudo apt install inetutils-traceroute # version 2:2.2-2ubuntu0.1, or
sudo apt install traceroute          # version 1:2.1.0-2
```

- nslookup google.com
Performs a **DNS lookup**.
Shows the **IP address** of google.com and the DNS server used.

```
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ nslookup google.com
Server:          127.0.0.53
Address:         127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.77.142
Name:   google.com
Address: 2404:6800:4007:810::200e
```

- netstat -tuln
Displays all **listening TCP and UDP ports** on your machine.
Useful to see which services are running.

```
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ netstat -tuln
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 127.0.0.1:631           0.0.0.0:*                LISTEN
tcp        0      0 127.0.0.53:53           0.0.0.0:*                LISTEN
tcp6       0      0 :::1:631                :::*                    LISTEN
udp        0      0 0.0.0.0:5353            0.0.0.0:*
udp        0      0 0.0.0.0:55878           0.0.0.0:*
udp        0      0 127.0.0.53:53           0.0.0.0:*
udp6       0      0 :::5353                 :::*
udp6       0      0 :::57496                :::*
```

- curl ifconfig.me
Shows your machine's **public IP address** by sending an HTTP request.

```
mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$ curl ifconfig.me
27.5.190.103mahalakshmi@mahalakshmi-VirtualBox:~/devops_project$
```